Normal-subgroup restriction of an acquired subgroup inventory

Human Compression project: L054 component

This is an elementary reuse method, not a new subgroup classification. The input is an already acquired complete list of actual embedded subgroup representatives, with its original completeness and group-computation premises. Printed names or published row counts are not substitutes for that input.

Theorem 1. *Let G be finite, $K \lhd G$, and $\mathcal{R}$ a complete covering list of G-conjugacy classes of subgroups. Let T be a transversal for G/K. For each $H \in \mathcal{R}$ with $H \leq K$, form H^t for every $t \in T$. Discard candidates only after actual K-conjugacy to an earlier retained candidate is established. The resulting list is a complete irredundant K-subgroup inventory. Repetitions in $\mathcal{R}$ cause no problem.*

Proof. If $L \leq K$, choose $H \in \mathcal{R}$ and $g \in G$ with $L = H^g$. Normality gives $H \leq K$. Write $g = tk$ with $t \in T$, $k \in K$. Then $L = (H^t)^k$, so its K-class is represented among the candidates. A right transversal works too, since left and right cosets coincide. Every candidate lies in K. Removing it only after a positive conjugacy test preserves its class, and comparison with all earlier survivors makes the final list irredundant. □

Normality is essential: for nonnormal K, an ambient representative outside K can have a conjugate inside K. The executable checks actual normality, input containment and transversal cardinality/coset distinctness. It retains the actual generating permutations and an origin (H, t) for each result. These checks do not replace the acquired completeness of $\mathcal{R}$. Class lengths in K are $[K : N_K(H)]$.

Lemma 2. *Suppose $[G : K] = 2$. An ambient class of subgroups lying in K consists of one or two K-classes. In the latter case every element outside K interchanges the two. The class of H does not split if and only if $G = KN_G(H)$.*

Proof. Normality makes conjugation by G act on K-conjugacy classes, and this action factors through G/K. The stabilizer of the class of H is $KN_G(H)$: if $H^g = H^k$, then $gk^{-1} \in N_G(H)$, and the converse is immediate. The assertions follow by the orbit–stabilizer formula. □

Theorem 3. *Assume $[G : K] = 2$, the ambient inventory is irredundant, and its weighted edges count all actual maximal subgroups in each fixed parent. For $H \leq K$ and an ambient child class C, let a be its maximal-subgroup multiplicity inside H. If C does not split, its multiplicity in the restricted inventory is a. If C splits but the parent class does not, each child class has multiplicity $a/2$. If both split, their separate multiplicities can instead be computed by the acquired robust local maximal-subgroup method in the actual parent. These rules give the complete restricted weighted maximal-incidence graph.*

Proof. All maximal subgroups of H lie in K, and maximality in H is independent of the ambient group. For a nonsplitting child class there is nothing to distribute. If the child splits and the parent does not, the lemma supplies $n \in N_G(H) \setminus K$. Conjugation by n permutes the actual maximal subgroups of H and exchanges the two child classes. It is a bijection between the two incidence fibres, proving equality and the evenness of a.

When both classes split, these arguments determine only their sum a. For any parent with such an ambiguous entry, the executable obtains a complete local maximal cover after checked

faithful degree reduction, deduplicates it by actual local conjugacy, and transports representatives back to H. Its direct local normalizer indices are added under actual K-conjugacy. This is exactly the acquired robust weight method. Every result must aggregate under its origin map to the acquired ambient edge row. Other parents use just the two inheritance rules already proved. Thus every actual maximal incidence is counted once. If c_H, c_L are class lengths in K and the forward weight is a_{HL}, double counting gives reverse weight $c_H a_{HL}/c_L$. The executable checks integrality and the two ambient class-size bounds. □

These edges are actual maximal inclusions, not covers of a conjugacy-class poset. No transitive reduction of the latter is used. Complete local covers, genuine maximality, exact membership/conjugacy/normalizers and faithful degree reduction remain explicit GAP interfaces as in the accepted atlas method. The first theorem needs a covering list; the weight-inheritance routine requires irredundancy and checks it in G.

Finite controls. Six tiny normal inclusions, with indices $1, 2, 4$, are checked both with the original complete ambient lists and with repetitions. These twelve controls produce 115 candidates and 46 retained classes in total and exercise actual splitting. Five index-two weight controls compare every entry against all actual subgroup inclusions, testing maximality by the absence of an actual intermediate subgroup. All 122 ordered pairs and 24 edges agree. The D_{16} to D_8 control uses two ambiguous-parent local computations; the other parents use inheritance. These are small controls, not atlas targets.

Planned application and limitations. The retained parent inventories for the current companion tables `pgammal216`, `psigmau34`, `pgammau35` and `pgammal225` contain possible normal index-two targets. The actual normal subgroup must still be identified with the intended named group by a defining action or exact transport. A source description, group order or matching numerical graph alone does not establish that identification. In particular the three index-two subgroups of the last parent must be distinguished without target row counts as a filter.

This component runs no atlas target and grants no new numerical or structural label credit. Current companion files are not asserted identical to historical online editions. The source attribution, whole-original admission, questions, formalization and compression measurements remain separate.